

1.) A CPU generally handles an interrupt by executing an interrupt service routine

[MCQ]

- A. As soon as an interrupt is raised
- B. By checking the interrupt register at the end of fetch cycle
- C. By checking the interrupt register after finishing the execution of the current instruction
- D. By checking the interrupt register at fixed time intervals

Explanation: - The CPU only checks for interrupts at specific points, typically after the completion of current instruction ensuring an orderly processing of interrupts.

2.) Which of the following devices should get higher priority on assigning interrupts? [MCQ]

- A. Hard disk
- B. Printer
- C. Keyboard
- D. Floppy disk

Explanation: - Hard-disk transfers large amount of data at high speed, requiring servicing to maintain system performance, hence they get higher priority over devices like printers, keyboards

3.) Which of the following statement is FALSE? [MCQ]

- A. In cycle stealing mode of DMA, one word of data is transferred between an I/O device and main memory in a stolen cycle
- B. For bulk data transfer, the burst mode of DMA has a higher throughput than the cycle stealing mode
- C. Programmed I/O mechanism has a better CPU utilization than the interrupt driven I/O mechanism
- D. none

Explanation: - Programmed I/O requires the CPU to continuously check the device status, wasting CPU cycles. Interrupt driven I/O is more efficient as it allows CPU to execute other tasks until an interrupt signals that the device is ready.

4.) In programmed I/O execution, the transfer time depends on _____? [MSQ]

- A. I/O speed
- B. Data Size
- C. Latency of I/O interface chip
- D. none of these

Explanation: - Transfer time is determined by the speed of the I/O device, the amount of data being transferred, and the latency in the communication interface.

5.) Which one of the following facilities transfer of bulk data from hard disk to main memory with the highest throughput? [MCQ]

- A. DMA based I/O transfer

- B. Interrupt driven I/O transfer
- C. Polling based I/O transfer
- D. Programmed I/O transfer

Explanation: - DMA allows data to be transferred directly between the hard-disk and memory without continuous CPU involvement, maximizing throughput.

6.)The primary function of an I/O interface is to:

- A) Convert data from machine format to human-readable format
- B) Act as a bridge between CPU and memory
- C) Manage communication between CPU and peripheral devices
- D) Store frequently used instructions

Explanation: -The I/O interface ensures proper data transfer between the CPU and peripherals by handling control signals and data exchange.

7) In programmed I/O, when can the CPU resume its execution of other instructions?

- A) Immediately after issuing an I/O request
- B) Only after the I/O operation is complete
- C) After an interrupt from the device
- D) After DMA completes the data transfer

Explanation: In programmed I/O, the CPU actively waits for the operation to complete, meaning it cannot execute other instructions during this time.

8.) How does the CPU determine the status of an I/O device in programmed I/O?

- A.) By continuously polling the devices status register
- B.)By waiting for an interrupt signal
- C.)By using direct memory access(DMA)
- D.)By checking the system cache.

Explanation: Programmed I/O involves the CPU repeatedly checking the device status in a loop, leading to inefficient CPU utilization.

9.)A system transfers data at 2000KB/s. How much time is required to transfer 10KB of data? [data transfer question]

- A.)0.005 sec
- B.)0.01sec
- C.)0.05 sec
- D.)0.1sec

Explanation: $\text{Time} = \text{Data Size} / \text{Transfer Rate} = 10 \text{ KB} / 2000 \text{ KB/s} = 0.005 \text{ sec.}$

10) A CPU executes an I/O operation that takes 5ms per transfer. If it completes 200 transfers per second, what is the total CPU time spent on I/O?

- A) 0.5 sec
- B) 1 sec**
- C) 2 sec
- D) 5 sec

Explanation: - Total CPU time = Number of transfers $\times$ Time per transfer = $200 \times 5\text{ms} = 1000\text{ms} = 1 \text{ sec.}$

11.) Assuming a device operating on 20MBPS rate, the time needed to check status in programmed I/O if status register size is 1 byte?
[MCQ]

- A. 50ns**
- B. 25ns
- C. 30s
- D. 0.05ns

Explanation: -

Given Data:

- Transfer rate = **20 Mbps (Megabits per second)**
- Status register size = **1 Byte (8 bits)**

Total CPU time = Number of transfers $\times$ Time per transfer = $200 \times 5\text{ms} = 1000\text{ms} = 1 \text{ sec.}$

Thus, the time required to check the status is **50 nanoseconds.**

12.) How does programmed I/O affect CPU performance? [MCQ]

- A. Reduces CPU utilization
- B. Increases CPU utilization**
- C. Eliminates the need for a system bus
- D. None of the above

Explanation: Since programmed I/O requires the CPU to continuously check device status, it leads to high CPU utilization without allowing it to process other tasks efficiently.

13.) Which of the following is not one of the types of buses? [MCQ]

- A. Control bus

- B. Data bus
- C. Address bus
- D. Utility bus

Explanation: Control, data, and address buses are standard types in a computer system. "Utility bus" is not a recognized type.

14.) What happens if an I/O device is not ready when the CPU polls it in programmed I/O?
[MCQ]

- A. The CPU moves to next instruction
- B. The CPU waits in a loop, repeatedly checking the device status
- C. The CPU triggers an interrupt and moves on to other tasks
- D. The I/O operation is canceled automatically

Explanation: Programmed I/O operates by continuous polling, meaning the CPU remains stuck in a loop until the device becomes ready.

15.) A system is using programmed I/O transfers 64KB of data in 8ms. What is the data transfer rate in KB/s?

- A. 4000 KB/s
- B. 6000 KB/s
- C. 8000 KB/s
- D. 10000 KB/s

Explanation: Transfer rate = Data Size / Time = 64 KB / 0.008 sec = 8000 KB/s.